Cover letter: B2 R2 submission

To: Dr. Ingrid Vasiliu-Feltes, Frontiers in Blockchain **From:** Zach Zukowski (Tokenization Systems; zach@tokenization.systems) **Re:** Auditing Governance Concentration Beyond Token Allocation: A Live-Governance Study of 52 Token Protocols (Manuscript ID 1853465; SSRN 6599278) **Working title (5 words):** Governance Concentration Beyond Token Allocation **Title note:** The full title is now “Auditing Governance Concentration Beyond Token Allocation: A Live-Governance Study of 52 Token Protocols.” The main title now leads with the verb “Auditing,” which foregrounds the measurement method that is the paper’s central contribution. The subtitle is “A Live-Governance Study of 52 Token Protocols,” chosen to be thesis-forward: “live-governance” is the paper’s own analytic concept (governance concentration as a live institutional state measured after protocol-controlled-address correction, not a launch-time allocation artifact), and “52 Token Protocols” corrects the prior “DePIN and DeFi” undercount, since the sample spans DePIN, DeFi, L1/L2 infrastructure, and social tokens rather than two sectors alone. The five-word working title remains “Governance Concentration Beyond Token Allocation” exactly, preserving editorial-system continuity across rounds. The institutional-design-and-philosophical-interpretation thread has been moved to a companion paper (Zukowski, 2026e), allowing the title and abstract to lead directly with the audit contribution. **Date:** 2026-06-03 **Round:** R2 revision

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Dear Dr. Vasiliu-Feltes and Reviewers,

I am submitting the R2 revision of “Auditing Governance Concentration Beyond Token Allocation: A Live-Governance Study of 52 Token Protocols” addressing Reviewer 1’s four Round 2 issues and incorporating substantial strengthening across data, methodology, and manuscript architecture. Point-by-point responses to all four reviewer issues are documented in the accompanying response document (`2026-05-26_R2_responses_master.md`), which is structured for direct copy-paste into the Frontiers reviewer-response web form.

Manuscript identity change since R1

The manuscript has been reframed around its empirical-measurement contribution. The R1 manuscript developed a five-lens philosophical framework integrated with empirical results; the revised manuscript leads with the measurement contributions and routes the philosophical-interpretation thread to a companion paper. This reframing preserves every R1 statistical finding. What is different is the architecture: empirical contributions are now the spine, with normative interpretation moved to a companion paper that develops the Polanyian fourth-fictitious-commodity argument at the depth that thesis warrants.

The revised manuscript reports four empirical contributions (Section 5.6):

1. **Generalizable PCA exclusion methodology** (Section 3.8) - 133 protocol-controlled-address exclusions across 38 protocols (plus a 2026 exchange-custody completion audit of 64 centralized-exchange deposit wallets across 21 of those protocols; full decomposition in Section 3.8 and Supplementary File S13) correcting systematic HHI

inflation in prior holder-list studies (median inflation factor 2.3x; maximum approximately 18x at RENDER). The five-class typology (burn-destination; foundation/treasury custody; staking-aggregation contracts; bridge custody and migration addresses; centralized exchange custody) generalizes to any cross-protocol governance HHI analysis on ERC-20 or SPL token holders.

2. **Allocation-null with insider-retention contrast** (Section 4.4) - Initial team-and-investor allocation does not explain post-distribution concentration (Pearson $r = 0.10$, $p = 0.49$, $N = 50$; statistically equivalent to zero within the powered envelope, two-one-sided-tests $p = 0.02$ within $|r| = 0.38$); current insider wallet presence among top-10 holders is associated with concentration (Spearman rho = 0.44, $p = 0.005$, $N = 39$; the count-based non-insider HHI tautology check at rho = 0.54, $p = 0.001$, $N = 34$ supports this as more than an arithmetic artifact).

3. **Corrected sector contrast** (Section 4.3) - DePIN governance concentration exceeds DeFi after protocol-controlled-address correction, a directionally robust medium effect. Under the headline voter-inclusive staking pass-through treatment, Cohen’s $d = 0.65$ (Mann-Whitney $p = 0.028$, N = 15/15; significant in all 30 leave-one-out iterations), with a marginal mean-based permutation (p approximately 0.08) that reflects a single concentrated DeFi-side vote-escrow bloc rather than an absence of effect, so the rank-based Mann-Whitney is the primary inference. A uniform staking-aggregation exclusion gives a comparable medium effect significant on every test (Cohen’s $d = 0.75$, Mann-Whitney $p = 0.018$, permutation $p = 0.009$), and the powered multivariate model keeps the DePIN coefficient significant (+0.87, $p = 0.02$, $N = 44$).

4. **Predominant delegation amplification with five structural exceptions** (Section 4.5) - Voting-HHI exceeds post-exclusion holding HHI in thirteen of eighteen protocols with sufficient governance data (range 2.5x Gnosis to 25.6x Polkadot; mean approximately 6.8x, median 4.4x). Five protocols disperse voting power below the holding baseline (ENS at 0.48x; GMX at 0.87x; HNT at 0.26x to 0.39x; JUP at 0.12x, the most-extreme dispersion outlier in the cross-section; and LPT at 0.27x, the most pronounced DePIN governance disperser, with Livepeer holding concentration unchanged at the cross-section maximum). Curve veCRV, Balancer veBAL, and Frax veFXS form a separate vote-escrowed class with extreme amplification (15x, 21x, and 11.4x respectively).

Robustness strengthening in this revision

Beyond the manuscript-vs-data reconciliation, the revision adds a zero-new-data robustness layer reproducible from committed inputs in the replication package. The sector contrast is now reported as an honest effect-size triple rather than a single number: we lead with the voter-inclusive staking pass-through treatment (Cohen’s $d = 0.65$, rank-significant at Mann-Whitney $p = 0.028$) and disclose its marginal mean-based permutation as the heavy-tail signature of one concentrated vote-escrow bloc rather than smoothing it, report the uniform staking-aggregation exclusion (Cohen’s $d = 0.75$) as significant on every test, and characterize the earlier large effect (Cohen’s $d = 1.05$) as inflated by an inconsistent staking treatment rather than as the headline. The allocation null is strengthened from a failure to reject into a positive bounded-effect result, equivalent to zero within the powered envelope (two-one-sided-tests $p = 0.02$ within $|r| = 0.38$) with the launch-design block jointly uninformative (omnibus $F(3,41) = 0.66$, $p = 0.58$). A descriptive temporal corroboration on the quarterly panel shows launch insider allocation uncorrelated with the 24-month governance-HHI endpoint (Pearson $r = -0.11$, $N = 13$), consistent with the cross-sectional null, and influence diagnostics confirm the sector contrast does not hinge on any single protocol (every powered-model leave-one-out refit keeps the DePIN coefficient significant). The subsidy-to-concentration association is characterized as demonstrated-fragile rather than absent: it is carried entirely by a single high-leverage DePIN observation (Livepeer), and relaxing the non-zero-subsidy filter to a larger $N = 35$ sample leaves the Livepeer-inclusive correlation essentially unchanged (Pearson $r = 0.61$) while the Livepeer-excluded association stays a clean null ($r = 0.09$, $N = 34$), so the larger cross-section confirms rather than rescues the fragility.

Reference additions since R1

The revision adds a small set of references that situate the measurement contribution within two established literatures it had previously drawn on only implicitly: the de jure versus de facto power distinction in comparative political economy, and the separation of ownership from control in corporate governance. These additions name the lineage behind framings the R1 manuscript already used but left uncited (for example, the description of outcomes as formally decentralized but substantively centralized), and they reinforce the audit-practice argument without altering any statistical finding. They are positioning citations for an empirical measurement paper, not a turn toward corporate-finance theory.

Submission package contents

- **Revised manuscript (clean version)** - the full R2 manuscript incorporating all changes described in the response document
- **2026-05-26_R2_responses_master.md** - Point-by-point response to Reviewer 1's R2 issues (markdown source for direct copy-paste into the Frontiers reviewer-response web form)
- **This cover letter** (PDF and DOCX)
- **Supplementary files**: S0 (statistics ledger); S1 (metric definitions); S2-S3 (codebook plus optional governance-scoring instruments retained as supplementary, not load-bearing for the empirical analysis); S4 (events table); S5 (empirical pipeline specification including per-iteration LOO data and per-protocol participation values); S6 (source provenance and revenue standardization); S7 (quarterly HHI panel for 14 governance tokens across 8 quarters); S8 (Token Terminal subsidy expansion analysis with sector-control multivariate models); S9 (Theil and Atkinson concentration-metric robustness check); S10 (PCA classification robustness across Specs A through E); S11 (Shapley-Shubik and Banzhaf power-index calculations); S12 (PCA-symmetric voting-HHI and signer-side functional PCA cases); S13 (Solana PCA exclusion audit including cross-protocol candidate-address verification); S14 (Shapley-Shubik and Banzhaf power-index full closure across the N=16 cross-source voting sample); S15 (voting-HHI coverage across the N=52 cross-section); S16a (Aethir, IoTeX, and ENS sensitivity analyses); S16b (additional protocols: FXS, SNX, GNO, TAO); S16c (Polkadot governance analysis); S17 (falsifiable predictions and pre-registration specification); S18 (EVM PCA classification with two-layer audit); S19 (Polkadot validator-set five-axis attribution methodology); S20 (Polkadot Subscan delegate-pool analysis); S21 (Class 3 versus Class

5 PCA disambiguation via transaction-pattern signatures); S22 (insider classification and staking-attribution audit); S23 (Q1-to-Q8 governance-HHI trajectory analysis on the 14-protocol panel); S24 (Optimism worked example); S25 (cross-protocol institutional concentration: voter-axis institutional-delegate audit plus holder-axis and validator-set CEX-overlap synthesis)

The address-by-address PCA exclusion log and the full regression dataset are available at the linked GitHub replication repository.

Methodology disclosure

The revision reconciles all reported statistics against a single source-of-truth regression dataset to prevent recurrence of the manuscript-vs-data drift Reviewer 1 surfaced in the R1 review. A statistics ledger (Supplementary File S0) lists every body-text statistic with its sample size and the reporting convention; the ledger is the single source of truth that body, tables, figures, captions, abstract, and conclusion reconcile against, and all reported statistics are reconciled against it. Snapshot-date discipline is acknowledged: Table 6 voting-HHI values use a March 2026 snapshot where available, consistent with the holder-list snapshot; the GMX and ENS additions use a May 2026 Tally snapshot, documented in the Table 6 footnote as a deliberate methodological exception for added protocols. The May 2026 Tally and Snapshot replication confirms delegate-pool drift; rank ordering across protocols is the more durable inferential property than absolute magnitude. To demonstrate that the 12-month voting-HHI values are not single-snapshot artifacts, the revision adds a full-history robustness companion (Table 6b) that pools each protocol's complete same-surface proposal history (the 12-month and full-history values are rank-stable at Spearman rho = 0.87 across the twelve protocols with a same-surface pool, with every amplify-or-disperse classification preserved) and a per-period voting-HHI trajectory figure (Figure 5) for the three protocols with a true governance series, confirming that voting concentration is stable across multi-year windows of normal participation.

Conflict of interest

The author served as Senior Investment Analyst at Borderless Capital, a cryptocurrency-focused investment firm, through February 2026 and retains no decision-making role, carried interest, or position-dependent compensation with the firm. The author may hold small personal positions in some of the protocols analyzed; positions were established outside the research period and are not position-dependent on Borderless Capital. A complete disclosure is provided in the Conflict of Interest statement accompanying this submission. No co-authors or other conflicts of interest.

Acknowledgment

We thank Reviewer 1 for the comprehensive cross-surface audit in R1 that surfaced the manuscript-vs-data drift class and motivated the reproduce-and-sync discipline adopted in this revision. Responding to that audit directly produced the substantive thesis evolution in the revised manuscript: the headline finding moved from "delegation amplifies unevenly with two infrastructure protocols showing opposite patterns" to "predominant amplification across thirteen of eighteen protocols with five structural exceptions documenting design-tractable dispersion." The revised manuscript's empirical contributions are stronger than the R1 manuscript's because

the reviewer’s audit framing was applied throughout the data, methodology, and architecture. We thank Reviewer 2 for the endorsement of publication.

Sincerely, Zach Zukowski Tokenization Systems zach@tokenization.systems ORCID: 0009-0006-3642-2450